

Fifteen Years of Work

ARCHIVES EXPRESS

The COMS Outcomes for Medium-Sized Choroidal Melanoma

CHOROIDAL MELANOMA is the most common primary ocular malignancy, with some 1500 new cases reported yearly in the United States. These tumors are both sight- and life-threatening, with approximately a 10% to 30% five-year mortality rate owing to systemic metastasis, depending on tumor size. Despite the major morbidity and mortality from these tumors, the condition occurs infrequently enough in any single ophthalmic practice of considerable size that clinical expertise in the condition had been difficult to accumulate in years past. Even the accurate diagnosis of choroidal melanoma had been sufficiently difficult to make, and an unfortunate fraction of eyes enucleated were found to have some other condition.

See also pages 951 and 969

The most appropriate treatment for choroidal melanoma was also uncertain. Enucleation was the standard treatment of the time when Zimmerman and McLean et al^{1,2} raised a question in the late 1970s as to whether enucleation might cause tumor dissemination and thereby increase the risk of mortality. Subsequent discussions of this hypothesis made it clear that more information was needed to determine appropriate treatment options. Although the urgency of the Zimmerman hypothesis highlighted the need for long-term follow-up studies comparing treatment alternatives, the relative infrequency of the condition made conduct of a study conceptually difficult. The National Eye Institute in 1983 and 1984 sponsored several meetings of ophthalmologists and oncologists, the outcome of which was a prospective multicenter clinical trial to judge treatment efficacy in the context of mortality, and the National Eye Institute then joined with the National Cancer Institute to initiate and support the Collaborative Ocular Melanoma Study (COMS).

The COMS addressed 2 primary questions involving mortality risks in the treatment of large- and medium-sized melanomas. In both cases, tumor irradiation was a component in 1 arm of the study. The COMS trial for large choroidal melanomas addressed the possible benefit of external beam radiotherapy prior to enucleation. The COMS trial for medium choroidal melanomas compared iodine 125 brachytherapy—by an episcleral radioactive plaque affixed to the eye-wall over the tumor—with enucleation as the alternative. In both cases the primary outcome involved mortality rates during long-term follow-up. Because numerous patients with choroidal melanoma experience other malignancies or die from other causes, the study design incorporated a Mor-

tality Coding Committee, which carefully assessed the cause of death in each case to judge whether or not this was related to the underlying choroidal melanoma.

Initial mortality findings in the COMS trial for medium-sized choroidal melanomas are reported in this issue of the ARCHIVES.³ This is a landmark study of long duration. Patient accrual ran 11 years, from 1987 to 1998, and these patients have now been followed up for 2 to 13 years. Great credit needs to be given to the dedication of our colleagues who accomplished the study. One must marvel at their collective exemplary bedside manner, as they needed to gain the confidence of their patients while explaining that a flip of the coin would determine which of 2 radically different treatments they would receive: removal of the eye vs potential salvage of the eye by application of a radioactive plaque, all the while with the clear specter of death over the shoulder. Clearly, to participate in a life-altering randomization event, these study patients had to have profound respect for their physicians as well as for the scientific and medical research trial process.

Before considering the specific outcomes of therapeutic efficacy, we should first turn to COMS report No. 17, also in this issue of the ARCHIVES,⁴ which analyzes the characteristics of the patients who enrolled. This report should be scrutinized carefully, as the applicability of the findings of any prospective randomized clinical trial depends on the extent to which enrolled patients are similar to those of the general population. The 1317 study patients were winnowed from nearly 9000 patients with choroidal melanomas who were evaluated at COMS centers, of whom about 5000 had tumors of eligible size. Patients with tumors adjacent to the optic disc were excluded, as this would impede surgical application of the radioactive plaque. Patients with predominantly ciliary body melanoma were also excluded. The presence of an additional primary cancer was also a cause for exclusion, as this would complicate attribution of mortality to the choroidal melanoma. These exclusions must be kept in mind when advising our patients, as a strict reading of the COMS outcome would stipulate some uncertainty with respect to the applicability of the results to patients not meeting inclusion criteria.

Approximately one half of the eligible patients agreed to participate in the randomization to enucleation vs brachytherapy, and the bottom line is that all factors judged to be important in potentially affecting the outcome were well balanced in the 2 arms of the trial. This report merits careful reading by anyone interested in the mechanics of prospective randomized clinical trials.

During the 11-year enrollment period, an essentially equal number of patients were randomly assigned

to enucleation or ^{125}I brachytherapy, and only a few were subsequently deemed ineligible. One remarkable ancillary finding of the study is that only 2 eyes of the 660 eyes that were enucleated were found histologically not to have melanoma, attesting to the remarkable improvements in diagnostic acumen that have occurred over the recent decades and providing a glimpse of one facet in the evolution of the high quality of patient care in the ophthalmic community.

Of the more than 1300 patients enrolled, approximately 1000, or 81%, have now been followed up for at least 5 years, and 416, or 32%, for 10 or more years. A total of 364 patients died after randomization, with an essentially equal number in each study arm; 188 (28%) of 660 enucleation patients vs 176 (27%) of 657 patients in the brachytherapy arm. Five-year unadjusted survival rates were comparable for the 2 therapies, at slightly better than 80% for both. In other words, this study showed that ^{125}I brachytherapy conveyed as good a survival rate as enucleation, in both a statistical and practical sense. This is an important statement for our choroidal melanoma patients, whom we previously could not advise with much authority as to therapy risks and benefits for medium-sized tumors.

The ethical precondition for conducting any prospective randomized clinical trial is that the outcome is unknown, even though currents of popular opinion might deem otherwise. Fortunately, patients in this COMS trial who underwent enucleation did not suffer as a consequence. Five-year death rates with histopathologically confirmed melanoma metastasis was 11% for enucleation, which was not statistically different than the 9% for brachytherapy.

What does this now mean for our clinical practices? This study follows by 3 years the COMS report on large choroidal melanomas,⁵ showing no benefit of radiation prior to enucleation. In that case, neither benefit nor harm was found from preenucleation external beam radiation. The present COMS findings on medium-sized choroidal melanoma demonstrate that ^{125}I radioactive plaque treatment alone, without enucleation, does not increase mortality compared with enucleation. We have a clear answer for our patients: medium-sized choroidal melanomas can be treated and the eye and some vision preserved by application of ^{125}I brachytherapy without increasing the risk of mortality over the previously accepted standard therapy of enucleation. If, for some reason, our patient wishes enucleation, this study can also be read in reverse, that careful enucleation does not increase the chance of metastasis and death compared with brachytherapy.

This study was exemplary in enrolling a large cohort of patients into the 2 study arms, thereby allowing solid estimates of risk ratios and derivation of confidence intervals for the outcomes. While it is conceivable that further study of a still larger set of patients with choroidal melanoma could alter these results, this is most unlikely. Formally stated, the trial had greater than 80% power to conclude that neither treatment increased or decreased mortality risk by as much as 25% relative to the other. These initial data give the relative risk for mortality as 0.99 for the 2 treatments (95% confidence interval, 0.80-1.22). Since the survival advantages of brachytherapy vs enucleation seem comparable for medium-sized tumors, quality of life becomes

an important factor in decisions between treatment options.⁶ Although visual acuity may suffer after brachytherapy, this treatment is not cosmetically disfiguring, and the eye retains some vision.

Important further results will also come from the COMS trial, as by 2003 all remaining patients with medium-sized tumors will reach the 5-year follow-up point. While it is quite unlikely that a subsequent divergence of risk outcomes will occur, at least over a 5-year follow-up period, compiling these mortality outcomes, ultimately on 10-year survival rates, will provide further information to help our patients. Based on the current statistical confidence intervals developed from the COMS data set, we are unlikely to see a subsequent divergence of mortality between ^{125}I brachytherapy vs enucleation for medium-sized tumors.

What of the original question of Zimmerman and colleagues as to whether enucleation might alter the natural history course of choroidal melanoma? Technically, the question is still not answered, as neither COMS trial compared enucleation with natural history alone. For medium-sized tumors, it is now unlikely that such a trial would be deemed ethical following the COMS outcome that the mortality risk with ^{125}I brachytherapy alone is the same as and no greater than enucleation. As Albert⁶ has pointed out, framing an appropriate study in which one arm is the natural history alone can be difficult, as the mere focusing attention on observation or intervention will probably bias the approach to the disease. Aggressive detection of tumors probably leads to earlier diagnosis and identifying smaller tumors. But one does not know whether small tumors are actually at an earlier stage or are less aggressive and carry more favorable risk. Studies of natural history of disease provide the best basis for subsequent therapeutic trials. All of this is a vindication of the rational scientific processes that were set into motion by the critical question originally posed by Zimmerman and colleagues. Although the basis of their hypothesis has not been substantiated, they performed a major service to all patients with choroidal melanoma in having this question addressed rigorously by a dedicated group of investigators.

Paul A. Sieving, MD, PhD
Ann Arbor, Mich

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